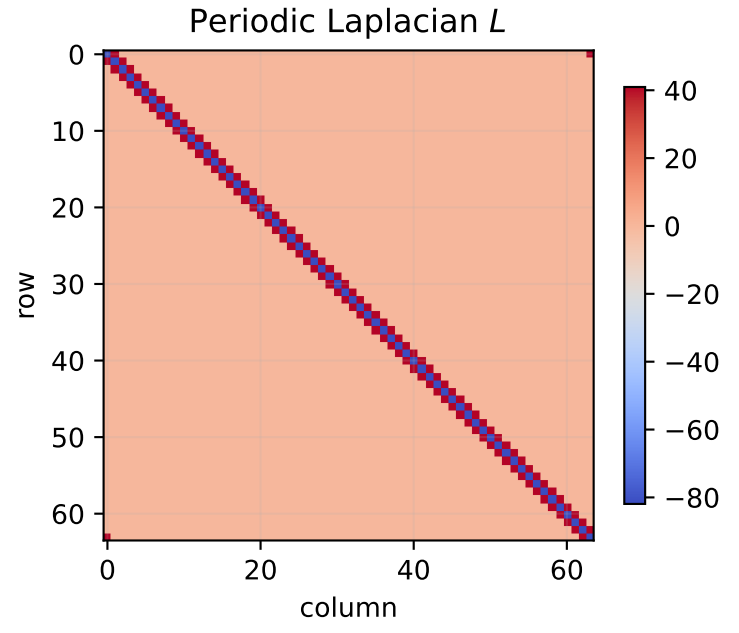
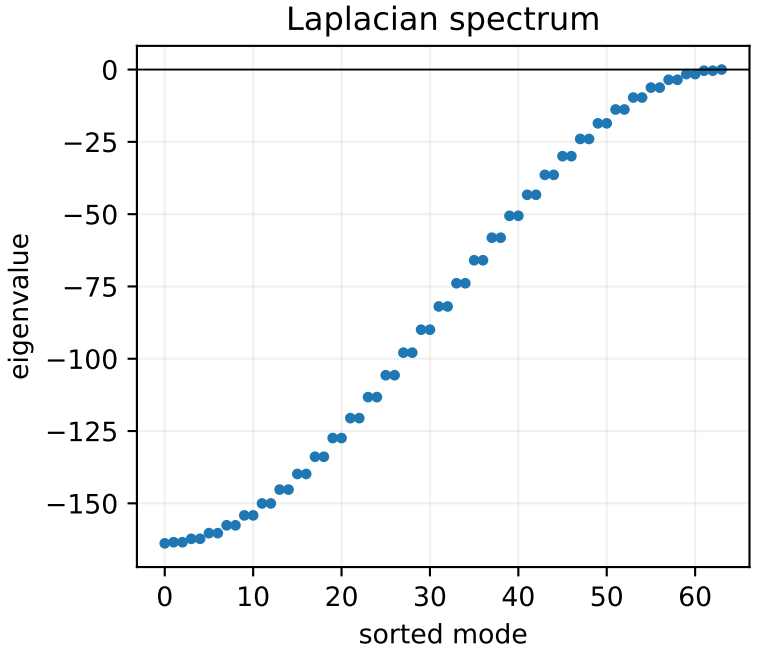
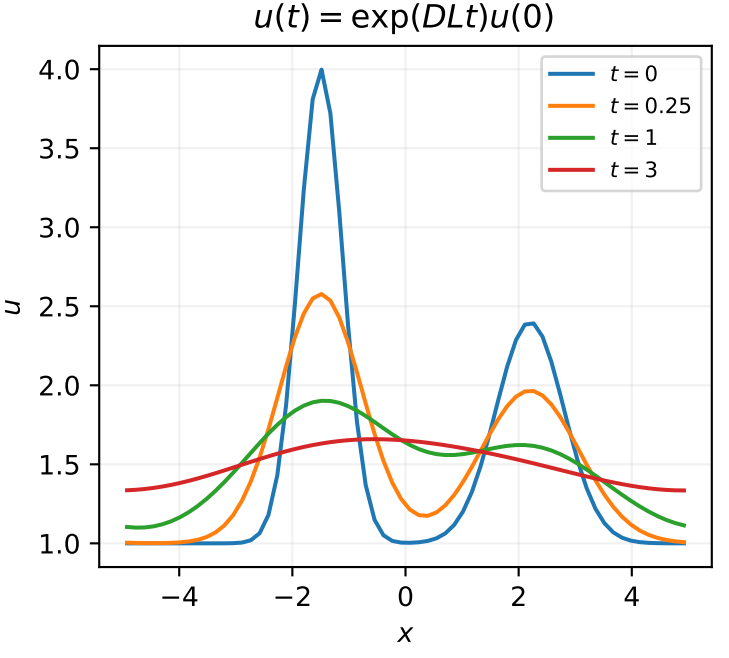
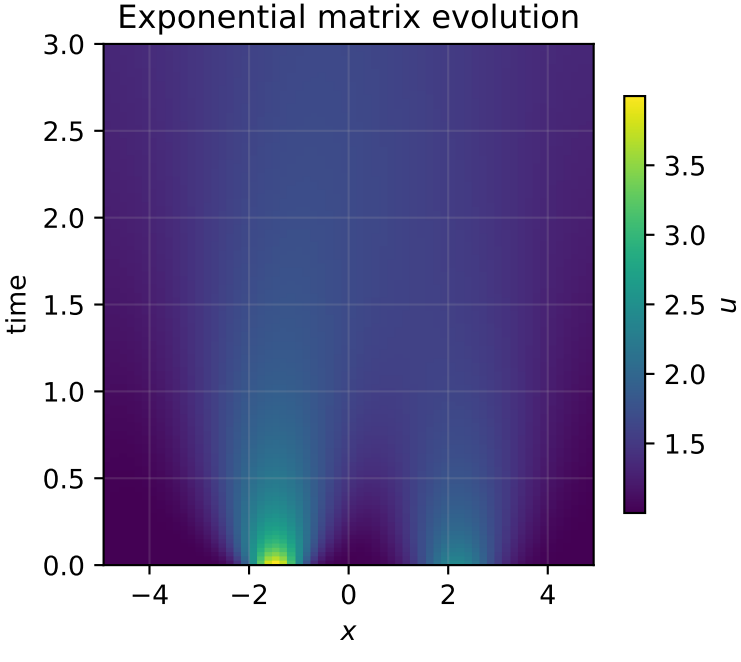
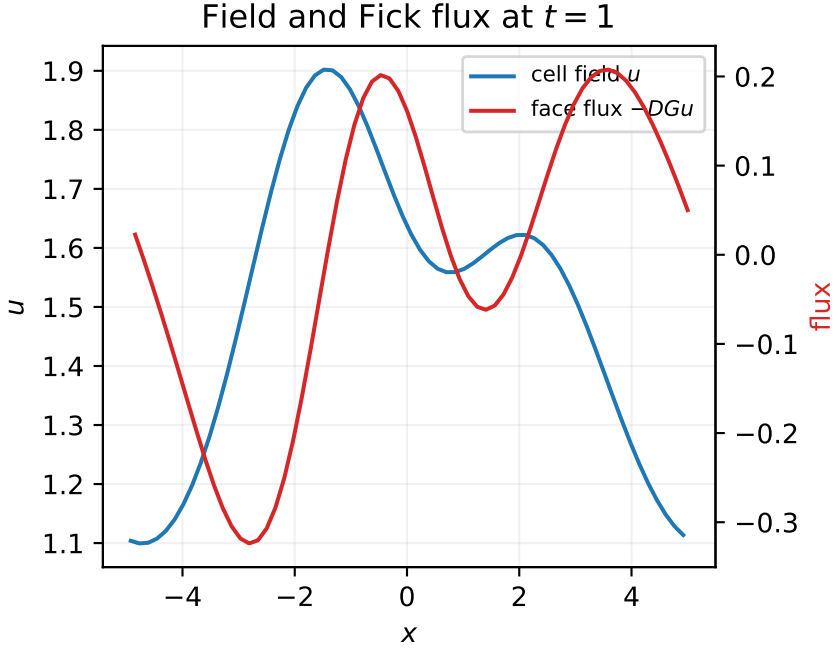
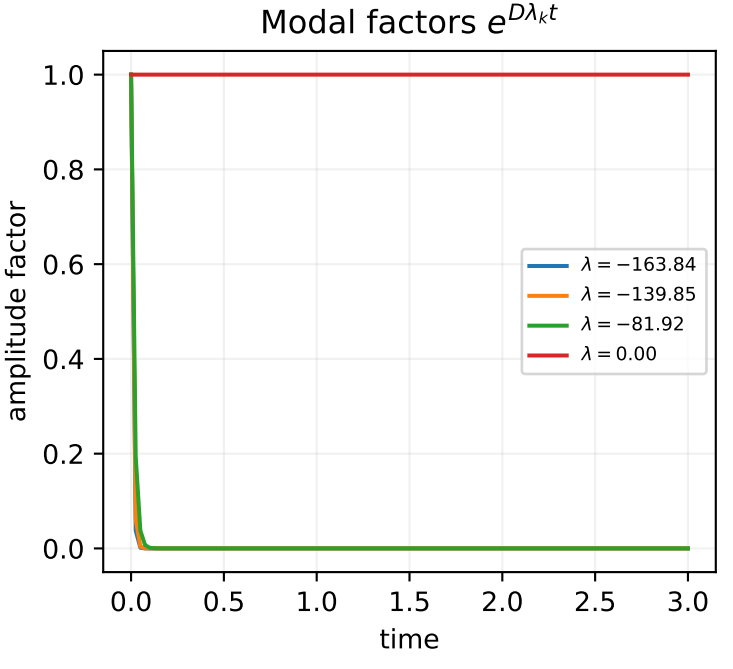

Periodic Laplacian L
row
column
0
10
20
30
40
50
60
40
20
0
−20
−40
−60
−80


Laplacian spectrum
eigenvalue
sorted mode
0
−25
−50
−75
−100
−125
−150
0
10
20
30
40
50
60


$u(t) = \exp(DLt)u(0)$
u
x
$t = 0$
$t = 0.25$
$t = 1$
$t = 3$
4.0
3.5
3.0
2.5
2.0
1.5
1.0
−4
−2
0
2
4


Exponential matrix evolution
time
x
u
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−4
−2
0
2
4
3.5
3.0
2.5
2.0
1.5


Field and Fick flux at $t = 1$
cell field u
face flux $-DGu$
u
flux
x
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
0.2
0.1
0.0
−0.1
−0.2
−0.3
−4
−2
0
2
4


Modal factors $e^{D\lambda_k t}$
amplitude factor
time
$\lambda = -163.84$
$\lambda = -139.85$
$\lambda = -81.92$
$\lambda = 0.00$
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0